

DOMINIC BELYAEV

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EDUCATION

Northwestern University | McCormick School of Engineering

Evanston, IL

B.S. in Mechanical Engineering - Concentration in Aerospace Engineering - GPA: 3.4

Expected 2028

Relevant Coursework: Mechanics of Materials, Thermodynamics, Stress Analysis, Fluid Mechanics, Materials Science, Electronics Design

WORK EXPERIENCE

Levanta Technologies

Diamondhead, MS

Mechanical Engineering Intern

Jun - Aug 2026

- Designed a passive retractable keel fin for a USAF/Navy-backed drone, rigid at 50 knots, folding 80° for platform recovery
- Ran Ansys Fluent CFD to correct drag coefficient from 0.10 to 0.011, a 9× cut in hinge moment allowing proper spring sizing
- Sized torsion springs with Shigley's method and iterated 3 printed prototypes in Fusion to final redundant twin-fin design
- Fabricated plugs, molds, & structural parts, and helped build a full-length carbon bottom skin reducing total weight by 5%

Meemo - Wristbuddy

Evanston, Illinois

Industrial Design Intern

Jun - Jul 2025

- Designed a 3-button actuation mechanism for a parent-child locating device, matching capped plungers to tactile PCB switches
- Packaged an RGB status indicator into the housing and verified plunger travel and component clearances in SOLIDWORKS

PROJECTS

Carbon Fiber Nosecone & Telemetry Electronics Rocket Build

May - Aug 2026

- Designed both part and tooling for a carbon nosecone: one-shot layup in a printed split female mold, 47% lighter than fiberglass
- Hand-soldered and integrated a Pico 2 flight computer (IMU + barometer) into the nose section, logging 100 Hz flight data
- Designed the avionics housing and recovery load path to a uniform SF ≈ 2 against the ~ 300 N parachute deployment spike
- Flew twice with full data capture & used the logs to trace root cause of flight 1's failure to parachute issues, then fixing & re-flying
- Directed an AI-assisted 3D reconstruction of both flights from raw IMU data, validated to 1.3 m against the onboard barometer

Northwestern University Space Technology & Rocketry Society (NUSTARS)

Jan 2026 - Present

- Designed a compact battery sled in Onshape for easy integration into ESRA IREC 25-26 competition rocket avionics bay
- Integrated sled to secure electronics against loads in ~ 7500 ft apogee flight using heat-set inserts and anti-vibration fasteners

NAR Level 1 Rocket Certification

Apr - Jun 2026

- Earned NAR L1 certification, owning the design, assembly, and launch of an H182-powered rocket with single-deploy recovery
- Modeled the rocket in OpenRocket to a 2,460 ft apogee and 1.6-cal stability, tuning ejection delay to the sim's 10.5 s optimum
- Assembled a 3-inch fiberglass airframe with epoxy, jig-aligning fins & reinforcing the roots with carbon-reinforced epoxy fillets

Bicycle Brake Caliper Design

Mar - Jun 2026

- Modeled & assembled arms in NX, validating each to a 1.2 safety factor against the 50 MPa Nylon-12 material stress limit
- Cut critical-arm peak stress by 16% and raised the deflection safety factor to 1.45 via an FEA & topology optimization redesign
- Ran on-bike braking tests per ISO 4210-2, beating the 7.62 m stopping-distance spec at 5.33 m after correcting a fit failure

DC Motor Control System

Jan - Mar 2026

- Built a dual-loop PID control system (1kHz current, 200Hz position) using C on a Raspberry Pi Pico for a brushed DC motor
- Achieved <25 mA average current tracking error and <9 degree average position tracking error through iterative PID tuning
- Developed a Python serial interface with 15+ real-time commands for trajectory generation, gain tuning, and data plotting

The Tele

Sep - Dec 2024

- Designed and prototyped an assistive diapering device enabling one-handed use for individuals with hemiplegia
- 3D printed custom handle with single-button telescoping locking pin mechanism, fabricated sidebars using common shop tools
- Conducted user testing with hemiplegia patients, iterating from an initial mockup to the final design based on usability feedback

MagClean

Mar - Jun 2025

- Designed and fabricated a magnet-powered whiteboard cleaning device using manual milling, band saw, drill press, and hand tools
- Conducted iterative user testing to optimize cleaning pressure, ergonomics, and reach for users of varying heights
- Reduced full-board cleaning time by $\sim 50\%$ vs. standard erasers & deployed in Northwestern's Ford Engineering Design Center

ACTIVITIES

Northwestern University Men's Rugby Club

Sep 2024 - Present

- 2+ hr practices twice a week, matches on weekends. Frequent out-of-state tournaments. Collegiate Division II rugby team.

SKILLS

Software: SOLIDWORKS, NX, Onshape, Fusion, Ansys, Python, C/C++, Raspberry Pi, Arduino, Microsoft 365 Suite, MATLAB

Design & Analysis: FEA, CFD, CAD, Topology Optimization, Stress Analysis, Rapid Prototyping, Structures Design, Electronics Design

Fabrication: 3D Printing, Composites Manufacturing, Lathe, Mill, Laser Cutter, Sheet Metal Tools, Bandsaws, Soldering